

**MATH 2101 LINEAR ALGEBRA I, FALL 2025 – OUTLINE OF
SOLUTION FOR ASSIGNMENT 3**

Due date: 2 October (Thursday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Solve the following system linear equations by Jordan-Gaussian eliminations:

$$\begin{aligned} -x_1 + 2x_2 + x_3 - x_4 + 3x_5 &= 7 \\ x_1 - 4x_2 + 5x_4 - 5x_5 &= -7 \\ -2x_1 + 6x_2 - x_3 - 5x_4 + 7x_5 &= 6 \\ -x_1 - 4x_2 + 4x_3 + 11x_4 - 2x_5 &= 11 \end{aligned}$$

Sol. Consider the augmented matrix associated to the above system:

$$(0.1) \quad \left(\begin{array}{ccccc|c} -1 & 2 & 1 & -1 & 3 & 7 \\ 1 & -4 & 0 & 5 & -5 & -7 \\ -2 & 6 & -1 & -5 & 7 & 6 \\ -1 & -4 & 4 & 11 & -2 & 11 \end{array} \right).$$

We apply the Jordan-Gaussian elimination:

$$\begin{aligned} (0.1) &\xrightarrow{\substack{R_1+R_2 \rightarrow R_2 \\ -2R_1+R_3 \rightarrow R_3 \\ -R_1+R_4 \rightarrow R_4}} \left(\begin{array}{ccccc|c} -1 & 2 & 1 & -1 & 3 & 7 \\ 0 & -2 & 1 & 4 & -2 & 0 \\ 0 & 2 & -3 & -3 & 1 & -8 \\ 0 & -6 & 3 & 12 & -5 & 4 \end{array} \right) \xrightarrow{\substack{-R_1 \rightarrow R_1 \\ R_2+R_3 \rightarrow R_3 \\ R_4-3R_2 \rightarrow R_4}} \left(\begin{array}{ccccc|c} 1 & -2 & -1 & 1 & -3 & -7 \\ 0 & -2 & 1 & 4 & -2 & 0 \\ 0 & 0 & -2 & 1 & -1 & -8 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array} \right) \\ &\xrightarrow{R_1-R_2 \rightarrow R_1} \left(\begin{array}{ccccc|c} 1 & 0 & -2 & -3 & -1 & -7 \\ 0 & -2 & 1 & 4 & -2 & 0 \\ 0 & 0 & -2 & 1 & -1 & -8 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array} \right) \xrightarrow{R_1-R_3 \rightarrow R_1} \left(\begin{array}{ccccc|c} 1 & 0 & 0 & -4 & 0 & 1 \\ 0 & -2 & 1 & 4 & -2 & 0 \\ 0 & 0 & -2 & 1 & -1 & -8 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array} \right) \\ &\xrightarrow{\substack{R_2 \rightarrow -1/2 R_2 \\ R_3 \rightarrow -1/2 R_3}} \left(\begin{array}{ccccc|c} 1 & 0 & 0 & -4 & 0 & 1 \\ 0 & 1 & -1/2 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1/2 & 1/2 & 4 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array} \right) \xrightarrow{R_2+1/2 R_3 \rightarrow R_2} \left(\begin{array}{ccccc|c} 1 & 0 & 0 & -4 & 0 & 1 \\ 0 & 1 & 0 & -9/4 & 5/4 & 2 \\ 0 & 0 & 1 & -1/2 & 1/2 & 4 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array} \right) \\ &\xrightarrow{\substack{R_2-5/4 R_4 \rightarrow R_2 \\ R_3-1/2 R_4 \rightarrow R_3}} \left(\begin{array}{ccccc|c} 1 & 0 & 0 & -4 & 0 & 1 \\ 0 & 1 & 0 & -9/4 & 0 & -3 \\ 0 & 0 & 1 & -1/2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array} \right). \end{aligned}$$

Therefore, x_4 is a free variable and we set $x_4 = t$. Then, the solution set is

$$\left\{ \left(4t + 1, \frac{9}{4}t - 3, \frac{1}{2}t + 2, t, 4 \right)^T \in \mathbb{R}^5 : t \in \mathbb{R} \right\}.$$

2. Find the reduced row echelon form of the following matrix:

$$\begin{pmatrix} c & 0 & 2 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

for all the values of $c \in \mathbb{R}$.

Sol. By the cofactor expansion along the 4th row,

$$\begin{aligned} \det \begin{pmatrix} c & 0 & 2 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 0 & 0 & 0 & 2 \end{pmatrix} &= 2 \det \begin{pmatrix} c & 0 & 2 \\ 1 & 2 & 3 \\ 2 & 4 & 8 \end{pmatrix} \\ &= 2c \det \begin{pmatrix} 2 & 3 \\ 4 & 8 \end{pmatrix} + 2 \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \\ &= 2c(4) - 0 \\ &= 2c \end{aligned}$$

Case 1: $c \neq 0$. Then the determinant of the matrix is non-zero and so the matrix is invertible. Hence, the reduced row echelon form is I_4 .

Case 2: $c = 0$. We carry out the Jordan-Gaussian elimination.

$$\begin{aligned} \begin{pmatrix} 0 & 0 & 2 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 0 & 0 & 0 & 2 \end{pmatrix} &\xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 \\ 2 & 4 & 8 & 10 \\ 0 & 0 & 0 & 2 \end{pmatrix} \xrightarrow{-2R_1 + R_3 \rightarrow R_3} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix} \xrightarrow{\frac{1}{2}R_2 \rightarrow R_2} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix} \\ &\xrightarrow{-2R_2 + R_3 \rightarrow R_3} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\frac{1}{2}R_3 \rightarrow R_3} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ &\xrightarrow{R_1 - 4R_3 \rightarrow R_1, R_2 - R_3 \rightarrow R_2} \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 - 3R_2 \rightarrow R_1} \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

3. Let A be an $n \times m$ matrix and let b be a $n \times 1$ column vector. Suppose all the coefficients of A and B are integers. Determine if the following statements are true or false.

- If the system $Ax = b$ has m leading variables, then any solution for the system $Ax = b$ has rational entries.
- If the system $Ax = b$ has n leading variables, then any solution for the system Ab has rational entries.

If the statement is true, explain it. If the statement is false, give a counterexample.

Sol. I claim that (a) is true and (b) is false.

- (a) We embed the set of integers $\mathbb{Z}$ into the field of rational numbers $\mathbb{Q}$. Apply the Jordan-Gaussian elimination to the system $(A|b)$ within $\mathbb{Q}$, we obtain the RREF of the form

$$\left(\begin{array}{c|c} I_m & b_1 \\ \hline 0_{(n-m) \times m} & b_2 \end{array} \right)$$

where b_1 resp. b_2 are column vectors of the size m resp. $n - m$, with rational entries (remember we apply the elimination using rational numbers, try examples). If the system has a solution, then $b_2 = 0_{(n-m) \times 1}$. In this case, the system has a unique solution as all the variables are leading variables. The solution is precisely b_1 , in particular with rational entries.

- (b) We give a counterexample. Consider the system

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

which satisfies the condition of (b). However, one observes that $(1 - \sqrt{2}, \sqrt{2}, 1)^T$ is a solution with entries being non-rational.

4. Let A be an invertible square 3×3 matrix.

- (a) Show that A is equal to UPL for some lower triangular matrix L , some upper triangular matrix U and permutation matrix P . (See Assignment 1 for the definition of a permutation matrix.)
 (b) Is the expression UPL above unique?

Proof. (a) We show that we can find an invertible lower triangular matrix L and an invertible upper triangular matrix U such that $UAL = P$ for some permutation matrix P . Note that using Jordan-Gaussian elimination, we know that the following operations can be realized as matrix multiplication by some L on the right (operation (i) and (iii)) and some U on the left (operation (ii) and (iv)) (try examples!):

- (i) Add j -th column to i -th column for $i < j$;
- (ii) Add j -th row to i -th row for $i < j$;
- (iii) Scaling columns;
- (iv) Scaling rows.

Now that by applying the operation (i) and (iii), we can reduce the last row to the form of $(0, 0, 1)$, $(0, 1, 0)$, or $(1, 0, 0)$. Then, applying operations (ii) and (iv), we reduce to the case where A becomes in the form (for instance)

$$\begin{pmatrix} * & 0 & * \\ * & 0 & * \\ 0 & 1 & 0 \end{pmatrix}.$$

Consider the submatrix by deleting the last row and the middle column (as in the above example). We apply a similar method to the two-by-two (invertible) submatrix and obtain a permutation matrix.

(b) No. Consider the decomposition

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

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